



Smart Workshop Tool-Carrying and Human-Following Robot

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Table Content

Topic No

Problem Identification

01

Structural and 3D Design Overview

02

BOM (Bill of Materials)

03

Pin Configuration

04

Circuit Diagram

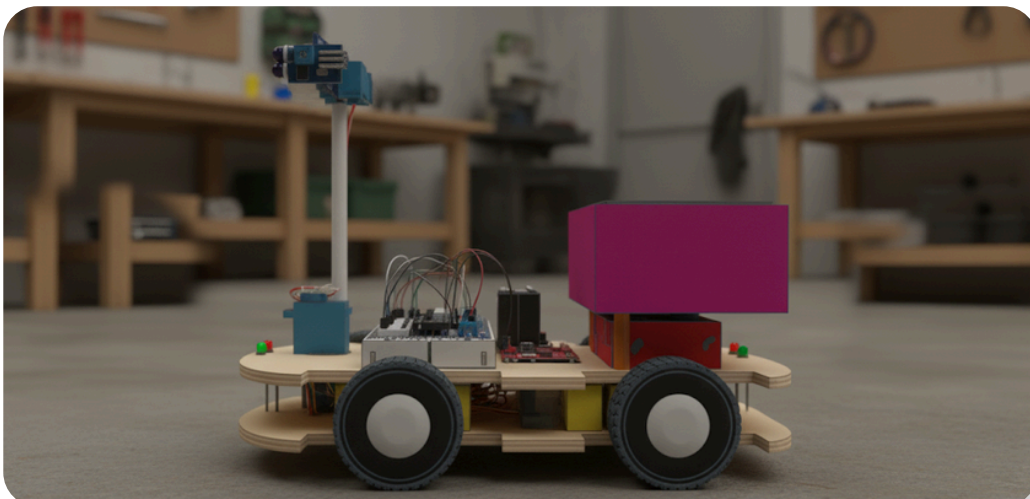
05

Flowchart

06

Project Timeline

07



01 Problem Identification

In modern workshops, technicians and engineers often need to move back and forth between different benches or tool racks to collect small tools, fasteners, and measuring instruments. While this may seem like a minor task, over the course of a workday, these repeated movements add up and create significant inefficiencies. The wasted time and physical strain not only slow down the work process but also reduce concentration and productivity.

Detailed Challenges Faced:

- **Wasted Time & Fatigue:** Technicians may walk dozens of times per day to retrieve or return tools. Each trip consumes time and contributes to physical exhaustion. This is especially problematic during long shifts or when multiple tool changes are required for a single task.
- **Workflow Disruption:** Continuous interruption breaks the worker's focus. For example, when measuring or assembling a device, stopping midway to fetch a missing screwdriver interrupts the workflow and increases the risk of mistakes.
- **Lack of Automation:** Current workshop setups typically rely on manual toolboxes or trays, which remain stationary. There are no affordable solutions that autonomously follow a worker around the workshop floor, keeping essential tools within arm's reach.

- **Safety Concerns:** Carrying tools by hand, especially sharp or heavy ones, can cause minor injuries or increase the chance of accidents. Workers often carry multiple items at once, further raising the risk of dropping tools or damaging components.
- **Reduced Efficiency in Group Settings:** In workshops with multiple technicians, tool-sharing or misplaced items are common. This adds further delays, as workers spend time searching for tools rather than focusing on their tasks.
- **Ergonomic Strain:** Repeated bending, lifting, or carrying small toolkits can cause back or shoulder strain over time. This is particularly challenging for older workers or those with physical limitations.
- **Environmental Hazards:** Workshops often involve welding, soldering, or working with chemicals that can produce sparks, flames, or smoke. These conditions may lead to accidental fires or poor air quality, which can affect worker health and safety.

Problem Definition and Proposed Solution :

Workshops today continue to face delays and inefficiencies because there is no dedicated robotic solution that is compact, safe, and affordable enough to assist technicians directly. A robotic system that can automatically follow the technician while carrying the required tools would greatly improve efficiency, safety, and convenience in such environments.

To enhance the safety of the working area, the robot is also equipped with additional smart sensors such as a Flame Detector and an Air Quality Sensor. These are used to identify potential hazards like fire or harmful gases. When a continuous flame is detected for 3–4 seconds, the robot stops, flashes both LEDs, and sends an alert to the user's phone through the Firebase real-time database system via Wi-Fi. This ensures reliable, instant message delivery to the paired mobile app. Similarly, if poor air quality or smoke is detected, the system sends a Firebase notification titled 'Hazardous Air Detected' and activates a visual warning. This short delay ensures that the robot does not react to normal welding sparks, which last only a second or two. Similarly, if poor air quality or smoke is detected, the system notifies the user with a "Hazardous Air Detected" message and activates a visual warning.

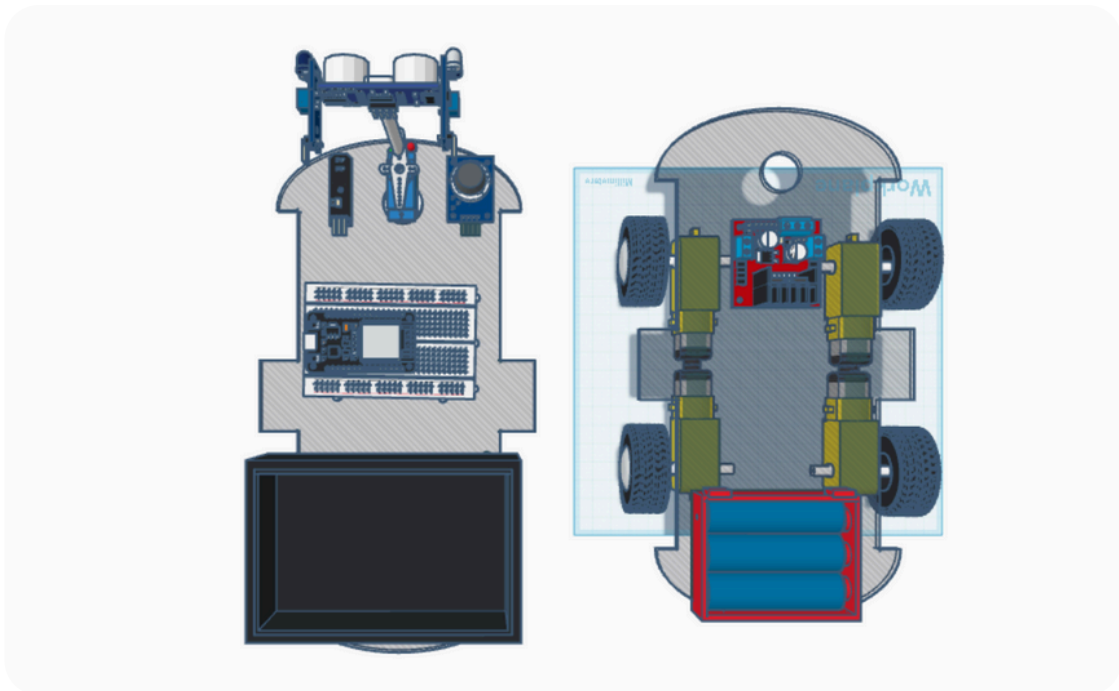
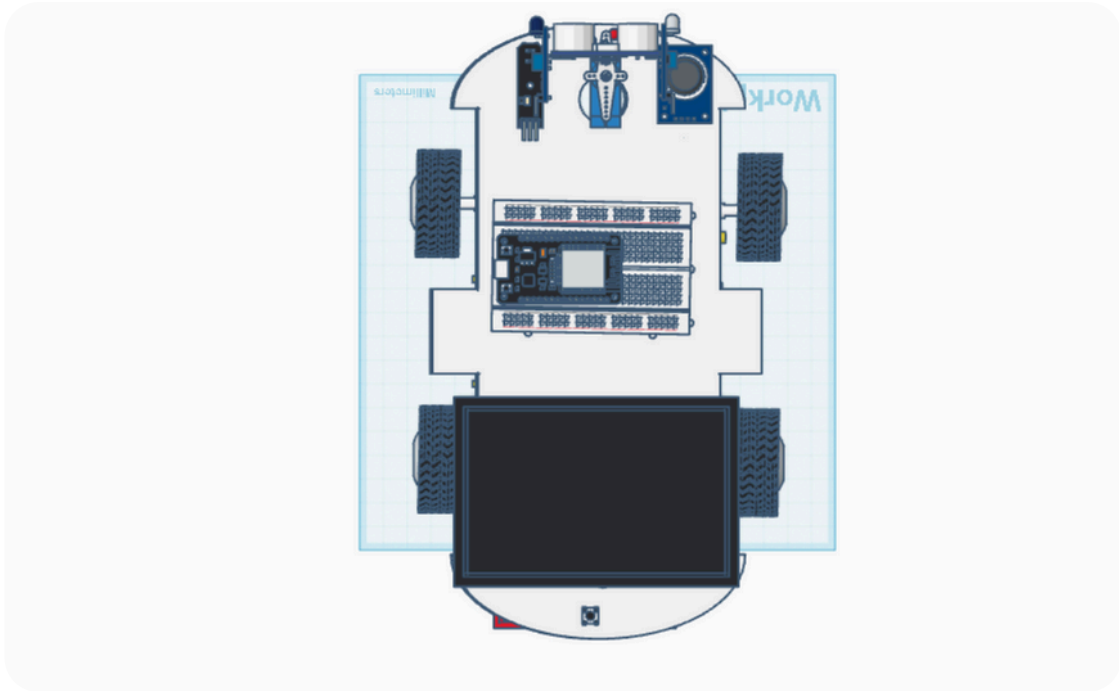
Such a system should:

- ***Follow only the paired user*** – By using Bluetooth RSSI, the robot recognizes the technician's phone and ensures it does not follow anyone else.

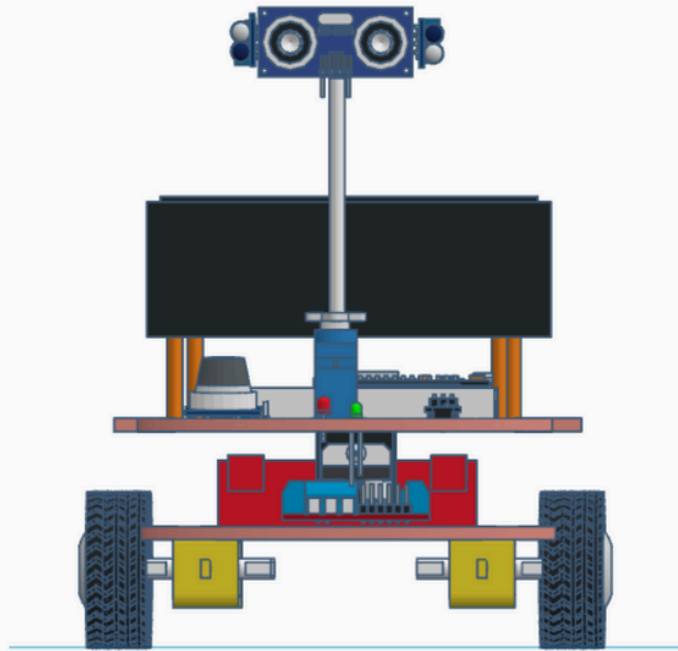
- **Maintain a safe distance (5–50cm)** – It remains close enough to be useful while stopping automatically at less than 5cm to prevent collisions.
- **Carry essential tools** – The robot provides a platform to transport the tools and equipment required for daily tasks, reducing repeated walking back and forth.
- **Notify the user clearly** – Through Wi-Fi or Bluetooth, it should send updates on status, such as “Connected – Following,” “Disconnected – Manual Mode,” “Searching...” or “Stopped – Too Close.”
- **Adapt to workshop environments** – It must move smoothly on workshop floors, detect and avoid obstacles, and handle minor surface unevenness without losing stability.
- **Reduce dependency on assistants** – By delivering tools directly, the robot eliminates the need for extra staff support, improving efficiency and saving time.

02 Structural and 3D Design Overview

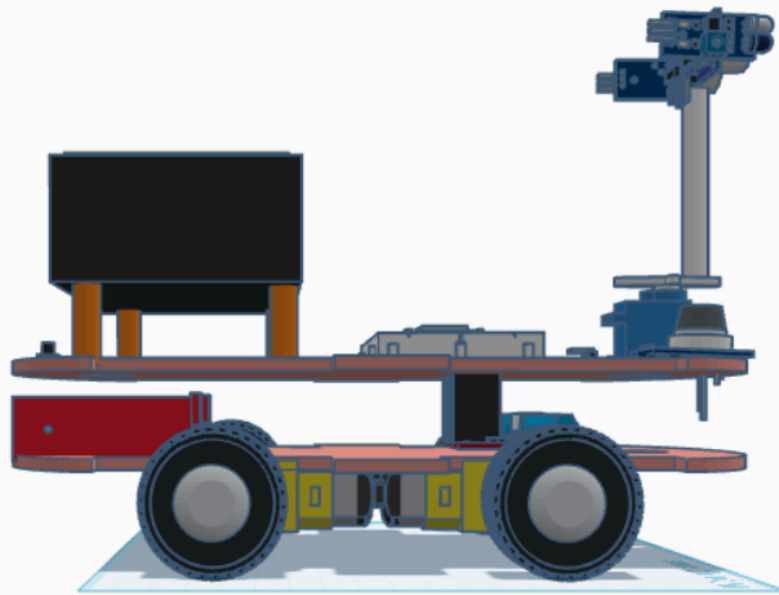
TOP VIEW



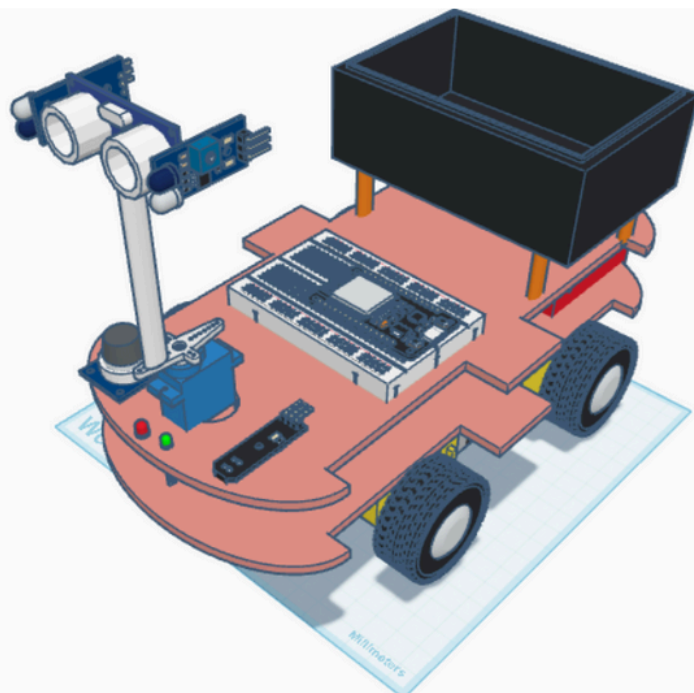
FRONT VIEW



SIDE VIEW



FRONT ANGLE VIEW



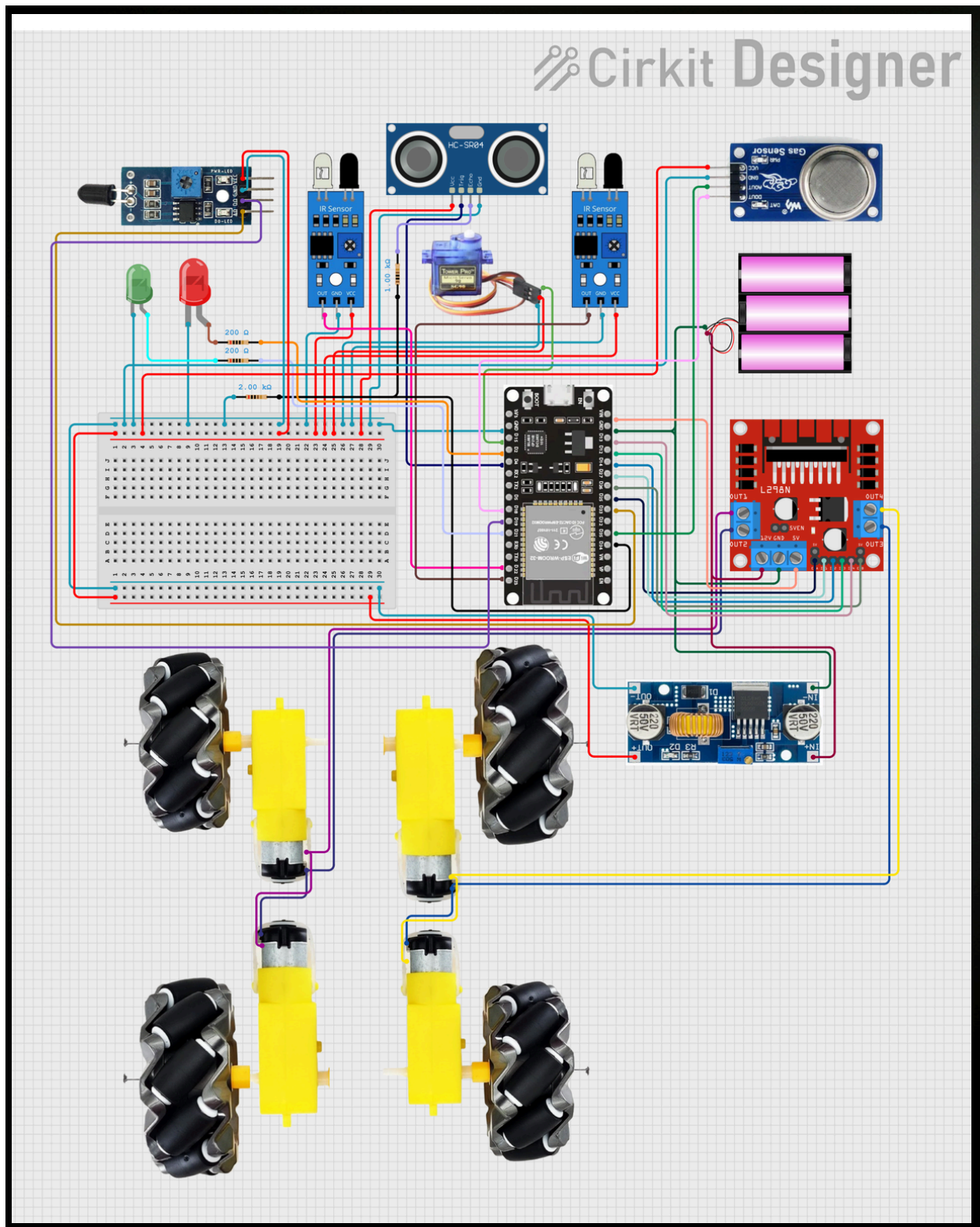
03 BOM (Bill of Materials)

Item No	Description	Units	Unit price	Total Price
Core Components				
1	ESP32S MODULE 30PIN	1	1100	1100
2	4WD CAR KITS	1	1550	1550
3	L298N MOTOR DRIVER	1	320	320
4	OBSTACLE SENSOR	3	80	240
5	ULTRASONIC SENSOR	3	190	570
6	18650 GREEN 1800MAH – FLAT TOP	3	295	885
7	18650X3 BATTERY HOLDER	1	85	85
8	DUAL BATTERY CHARGER UK PLUG	1	475	475
9	MALE TO MALE (10CM)	1	140	140
10	MALE TO FEMALE (30CM)	1	170	170
11	830TP BREADBOARD	1	175	175
12	5MM BLUE DEFUSED LED	3	2.5	7.5
13	5MM RED DEFUSED LED	3	2.5	7.5
14	SG90 SERVO MOTOR 180 DEGREES	1	240	240
15	ULTRASONIC SENSOR BRACKET	1	100	100
16	Flame Detector Infrared IR Module	1	170	170
17	MQ-135 GAS SENSOR	1	360	360
				6595
Tools				
1	7X300MM CLEAR GLUE STICK	5	20	100
2	20W GLUE GUN – NORMAL UK PLUG	1	395	395
3	40W GOOT SOLDERING IRON CHN – UK PLUG	1	680	680
4	Solder Wire (3m)	1	220	220
5	Soldering Iron (Adjustable Temp - 60W)	1	1725	1725
	Total Cost			9715
	Actual group cost			9715
	Cost per member (assuming 3 member group)			3238.333

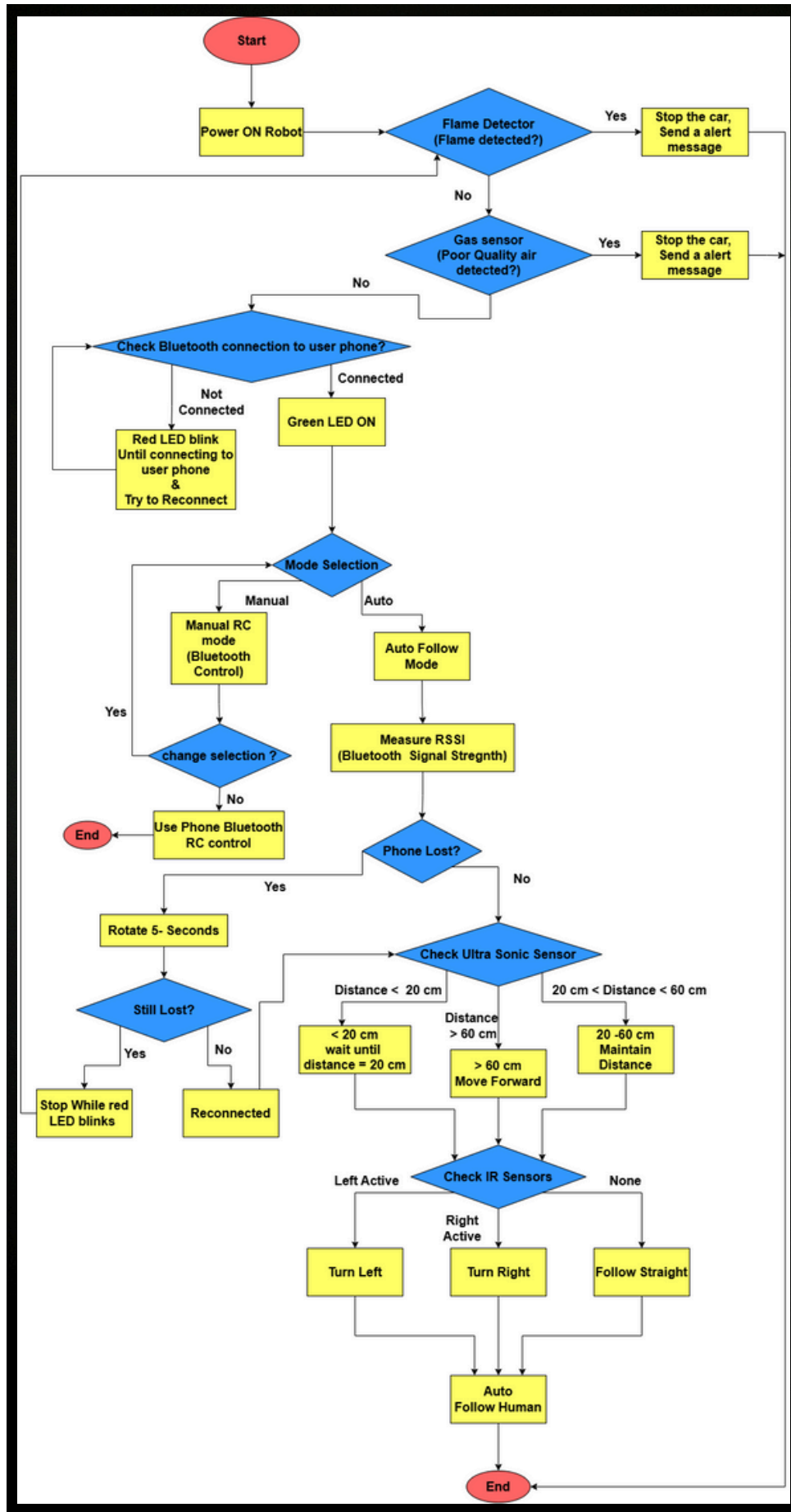
04 Pin Configuration

ESP32 Pin	Component	Description	Signal	Type(Analog/Digital)
GPIO25	L298N Motor Driver	Drives left motors speed (PWM)	ENA	Digital (PWM)
GPIO27	L298N Motor Driver	Sets left motors direction	IN1	Digital
GPIO14	L298N Motor Driver	Sets left motors direction	IN2	Digital
GPIO26	L298N Motor Driver	Drives right motors speed (PWM)	ENB	Digital (PWM)
GPIO12	L298N Motor Driver	Sets right motors direction	IN3	Digital
GPIO13	L298N Motor Driver	Sets right motors direction	IN4	Digital
GPIO4	HC-SR04 Ultrasonic (Front)	Distance sensing	TRIG	Digital
GPIO34	HC-SR04 Ultrasonic (Front)	Echo input via resistor divider	ECHO	Digital (Pulse Timing)
GPIO22	IR Sensor (Left)	Left/right hint for alignment	DO	Digital
GPIO23	IR Sensor (Right)	Left/right hint for alignment	DO	Digital
GPIO15	Servo Motor	Rotates sensor bracket to scan	SIG	Digital (PWM)
GPIO2	Status LED (Red)	Shows STOP/FAULT or <20 cm stop	LED+	Digital (PWM)
GPIO21	Status LED (Green)	Shows CONNECTED/FOLLOWING OK	LED+	Digital (PWM)
GPIO33	Flame Detector IR (MD0221)	Flame intensity analog read	AO	Analog
GPIO19	Flame Detector IR (MD0221)	Flame threshold detect	DO	Digital
GPIO35	MQ-135 Gas Sensor (MD0373)	Air quality analog read via divider	AO	Analog
GPIO18	MQ-135 Gas Sensor (MD0373)	Air quality threshold detect via divider	DO	Digital
	Motor Driver	LED		
	Ultrasonic Sensor	Flame Detector		
	IR Sensor	Gas Sensor		
	Servo Motor			

05 Circuit Diagram



06 Flow Chart

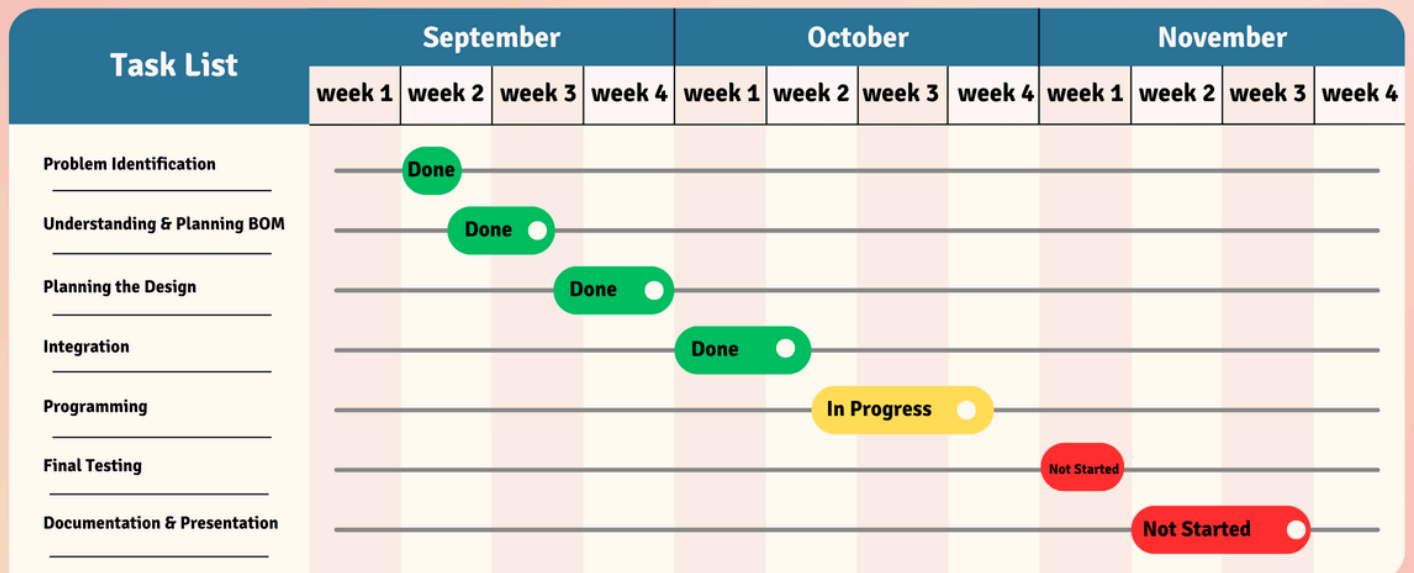


07 Project Timeline & Gantt Chart

Phase	Activity	Duration	Planned End Date	Actual Status	Actual Completion	Milestone / Deliverable
Phase 01	Problem Identification	3 Days	Sep 10, 2025	✓ Completed	11-Sep-25	Problem identified
Phase 02	Understanding & Planning BOM	7 Days	Sep 17, 2025	✓ Completed	17-Sep-25	BOM finalized and approved
Phase 03	Planning the Design	12 Days	Sep 29, 2025	✓ Completed	1-Oct-25	Completed 3D design, sketches, and flowcharts
Phase 04	Integration	10 Days	Oct 10, 2025	✓ Completed	17-Oct-25	Partial system integration underway
Phase 05	Programming	15 Days	Oct 26, 2025	🟡 In Progress	TBD	Code development planned
Phase 06	Final Testing	10 Days	Nov 06, 2025	⌛ Not Started	TBD	Verification & validation phase
Phase 07	Documentation & Presentation	14 Days	Nov 20, 2025	⌛ Not Started	TBD	Final report and presentation preparation

Project Timeline Gantt Chart

ROBOT
MAKING



Thank You